

Development and Validation of Mini-Bus Tracking System

College of Information and Computer Studies

Interworld Colleges Foundation Inc.

Burgos St., Paniqui, Tarlac

Robert Pabalate

Jericho Cariño

Henry Garcia

Jay Mark Baranda

Bachelor of Science in Computer Science

May 2026

Abstract

Robert Pabalate, Jericho Cariño, Henry Garcia, and Jay Mark Baranda. *Development and Validation of Mini-Bus Tracking System*. Bachelor of Science in Computer Science. Interworld Colleges Foundation Inc., Burgos St., Paniqui, Tarlac, Philippines.

Adviser: Beverly B. Pascua, MIT

Public transportation in many areas continues to face inefficiencies, such as unpredictable arrival times, overcrowding, and a lack of reliable information for commuters. These issues often discourage people from using public transport and hinder efforts to modernize the system. Advances in GPS and mobile technologies have shown promise in tackling these problems by providing real-time updates and better coordination between operators and passengers. In the Philippines, informal transport modes like minibuses remain mostly untracked, leaving passengers vulnerable to delays and service disruptions. To address these challenges, this study introduces the Mini-Bus Tracking System, a commuter-focused platform that offers real-time bus location and estimated arrival times, live passenger count monitoring, secure user registration and authentication, and verified personnel details for accountability. These features aim to boost commuter confidence, increase operational transparency, and support safer, more organized mobility. The main goal of this research is to develop and validate this system as a localized solution to improve the efficiency and reliability of public transportation.